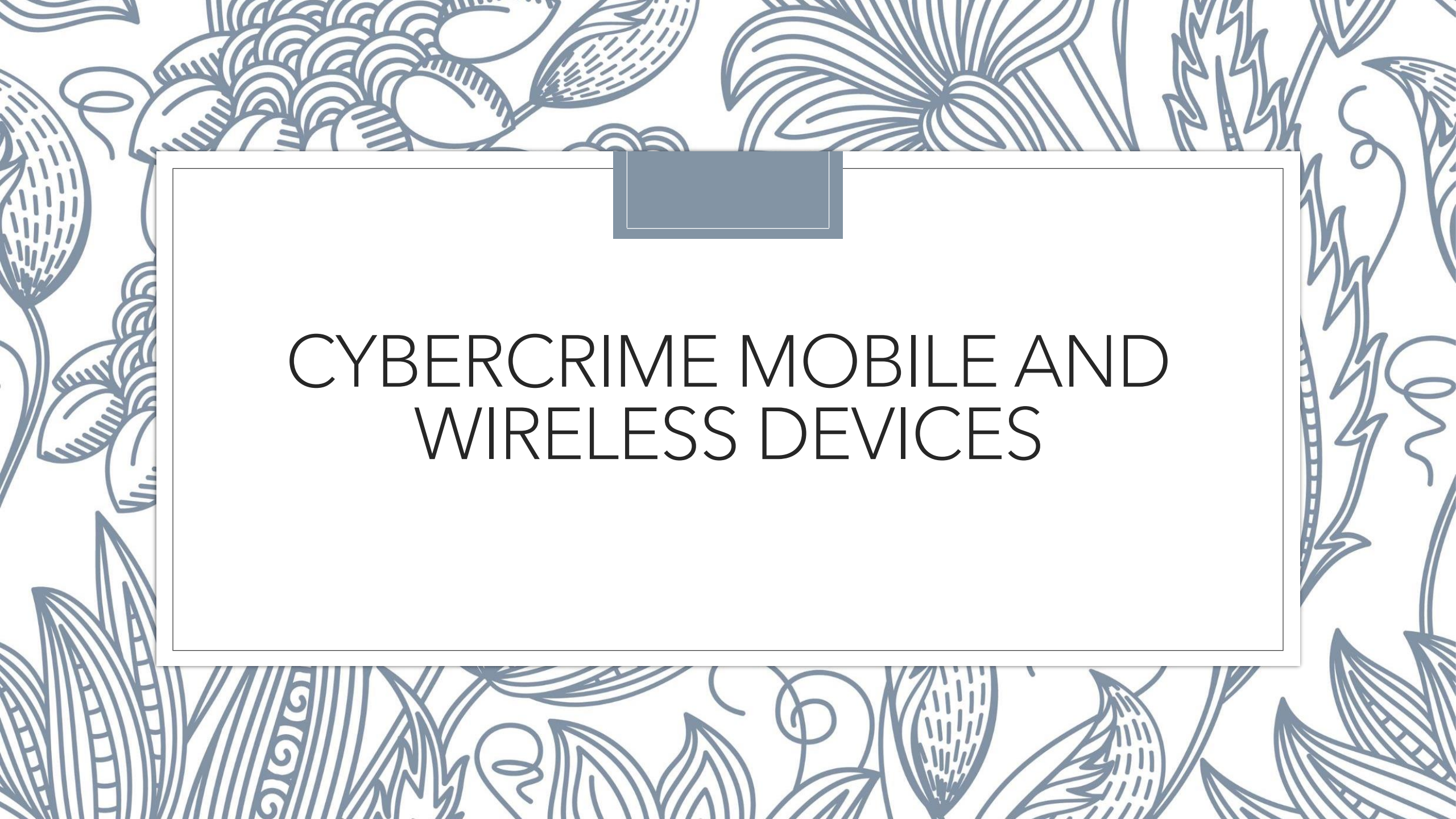




CYBERCRIME MOBILE AND WIRELESS DEVICES

Proliferation (Growth) of Mobile and Wireless Devices

- Today, incredible advances are being made for mobile devices.
- The trend is for smaller devices and more processing power.
- A few years ago, the choice was between a wireless phone and a simple PDA. Now the
- buyers have a choice between high-end PDAs with integrated wireless modems and small
- phones with wireless Web-browsing capabilities.
- A simple hand-held mobile device provides enough computing power to run small applications, play games and music, and make voice calls.
- As the term “mobile device” includes many products. We first provide a clear distinction among the key terms: mobile computing, wireless computing and hand-held devices.

Mobile computing

- Mobile computing is “taking a computer and all necessary files and software out into the field.”
- Many types of mobile computers have been introduced since 1990s.

They are as follows:

- 1. Portable computer:** It is *a general-purpose computer that can be easily moved from one place to another*, but cannot be used while in transit, usually because it requires some “setting-up” and an AC power source.
- 2. Tablet PC:** It lacks a keyboard, is shaped like a slate or a paper notebook and has features of a touch screen with a stylus and handwriting recognition software. Tablets may not be best suited for applications requiring a physical keyboard for typing, but are otherwise capable of carrying out most tasks that an ordinary laptop would be able to perform.
- 3. Internet tablet:** It is the Internet appliance in tablet form. Unlike a Tablet PC, the Internet tablet does not have much computing power and its applications suite is limited. Also it cannot replace a general-purpose computer. The Internet tablets typically feature an MP3 and video player, a Web browser, a chat application and a picture viewer.
- 4. Personal digital assistant (PDA):** It is a small, usually pocket-sized, computer with limited functionality. It is intended to supplement and synchronize with a desktop computer, giving access to contacts, address book, notes, E-Mail and other features.

5. Ultramobile PC: It is a full-featured, PDA-sized computer running a general-purpose operating system (OS).

6. Smartphone: It is a PDA with integrated cell phone functionality. Current Smartphones have a wide range of features and installable applications.

7. Carputer: It is a computing device installed in an automobile. It operates as a wireless computer, sound system, ***global positioning system (GPS)*** and **DVD player**. It also contains word processing software and is Bluetooth compatible.

8. Fly Fusion Pentop computer: *It is a computing device with the size and shape of a pen.* It functions as a writing utensil, MP3 player, language translator, digital storage device and calculator.

Wireless computing

Wireless refers to the method of transferring information between a computing device (such as a PDA) and a data source (such as an agency database server) without a physical connection.

□ Not all wireless communication technologies are mobile. For example, lasers are used in wireless data transfer between buildings, but cannot be used in mobile communications at this time.

□ Mobile simply describes a computing device that is not restricted to a desktop, that is, not tethered. As more personal devices find their way into the enterprise, corporations are realizing cybersecurity threats that come along with the benefits achieved with mobile solutions.

□ Mobile computing does not necessarily require wireless communication. In fact, it may not require communication among devices at all.

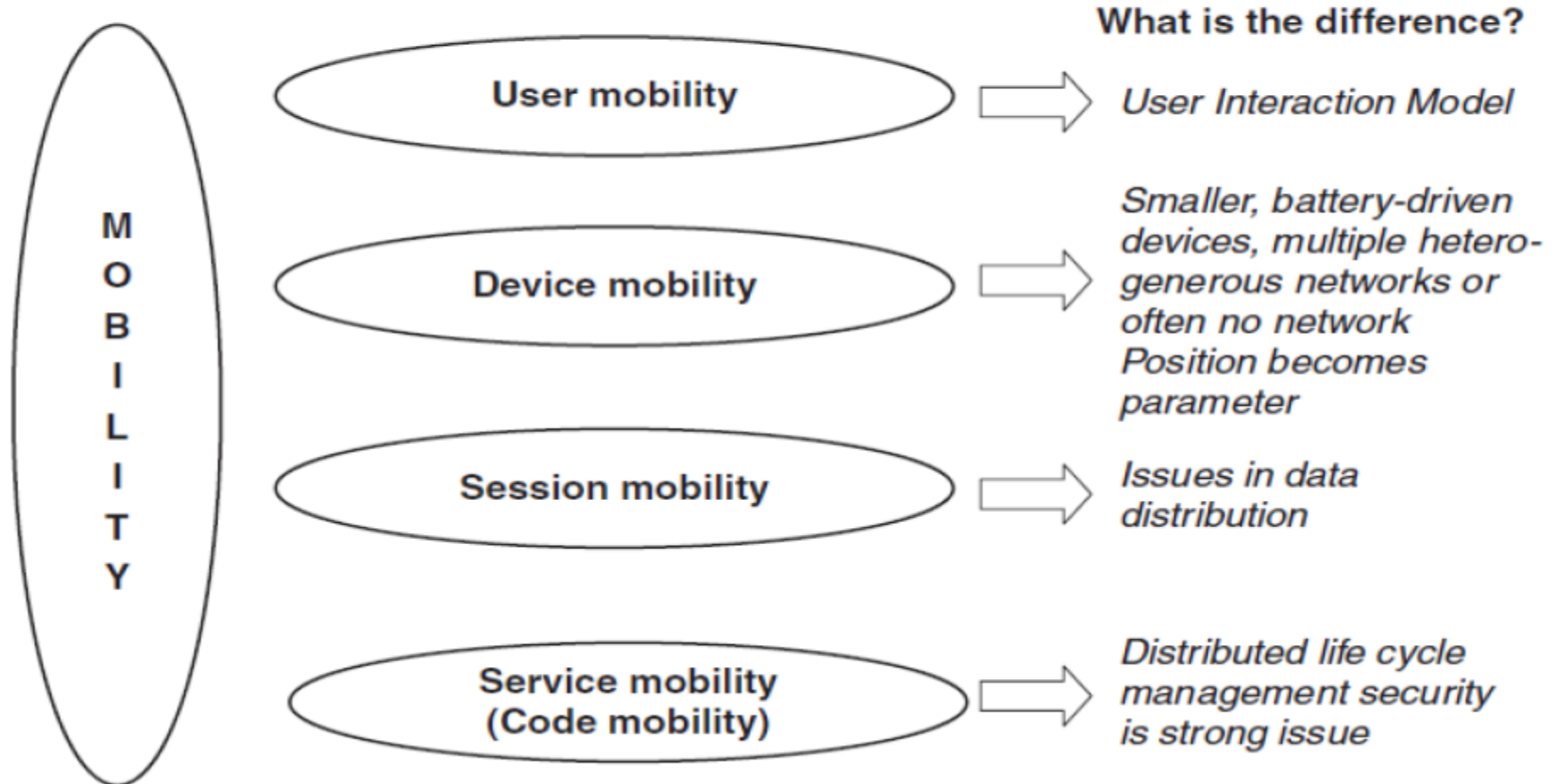
□ Thus, while “wireless” is a subset of “mobile,” in most cases, an application can be mobile without being wireless.

□ Smart hand-helds are defined as hand-held or pocket-sized devices that connect to a wireless or cellular network, and can have software installed on them; this includes networked PDAs and Smartphones

Trends in Mobility

- Mobile computing is moving into a new era, third generation (3G), which promises greater variety in applications and have highly improved usability as well as speedier networking.
- “iPhone” from Apple and Google-led “Android” phones are the best examples of this trend and there are plenty of other developments that point in this direction.
- This smart mobile technology is rapidly gaining popularity and the attackers (hackers and crackers) are among its biggest fans.
- It is worth noting the trends in mobile computing; this will help readers to realize the seriousness of cybersecurity issues in the mobile computing domain.

Types of Mobility and its Implications



Types of attacks against 3G mobile networks

Popular types of attacks against 3G mobile networks are as follows:

1. Malwares, viruses and worms: Although many users are still in the transient process of switching from 2G, 2.5G to 3G, it is a growing need to educate the community people and provide awareness of such threats that exist while using mobile devices. Here are few examples of malware(s) specific to mobile devices:

- Skull Trojan: It targets Series 60 phones equipped with the Symbian mobile OS.
- Cabir Worm: It is the first dedicated mobile-phone worm; infects phones running on Symbian OS and scans other mobile devices to send a copy of itself to the first vulnerable phone it finds through Bluetooth Wireless technology. The worst thing about this worm is that the source code for the Cabir-H and Cabir-I viruses is available online.
- Mosquito Trojan: It affects the Series 60 Smart phones and is a cracked version of "Mosquitos" mobile phone game.
- Brador Trojan: It affects the Windows CE OS by creating a svchost.exe file in the Windows start-up folder which allows full control of the device. This executable file is conducive to traditional worm propagation vector such as E-Mail file attachments
- Lasco Worm: It was released first in 2005 to target PDAs and mobile phones running the Symbian OS. Lasco is based on Cabir's source code and replicates over Bluetooth connection.

Attacks

- **Denial-of-service (DoS):** The main objective behind this attack is to make the system unavailable to the intended users. Virus attacks can be used to damage the system to make the system unavailable.
- **Overbilling attack:** Overbilling involves an attacker hijacking a subscriber's IP address and then using it (i.e., the connection) to initiate downloads that are not "Free downloads" or simply use it for his/her own purposes. In either case, the legitimate user is charged for the activity which the user did not conduct.
- **Spoofed policy development process (PDP):** These types of attacks exploit the vulnerabilities in the GTP [General Packet Radio Service (GPRS) Tunneling Protocol].
- **Signaling-level attacks:** The Session Initiation Protocol (SIP) is a signaling protocol used in IP multimedia subsystem (IMS) networks to provide Voice Over Internet Protocol (VoIP) services. There are several vulnerabilities with SIP-based VoIP systems.

Credit Card Frauds in Mobile and Wireless Computing Era

These are new trends in cybercrime that are coming up with mobile computing – mobile commerce (M-Commerce) and mobile banking (M-Banking).

□ Credit card frauds are now becoming commonplace given the ever-increasing power and the ever-reducing prices of the mobile hand-held devices, factors that result in easy availability of these gadgets to almost anyone.

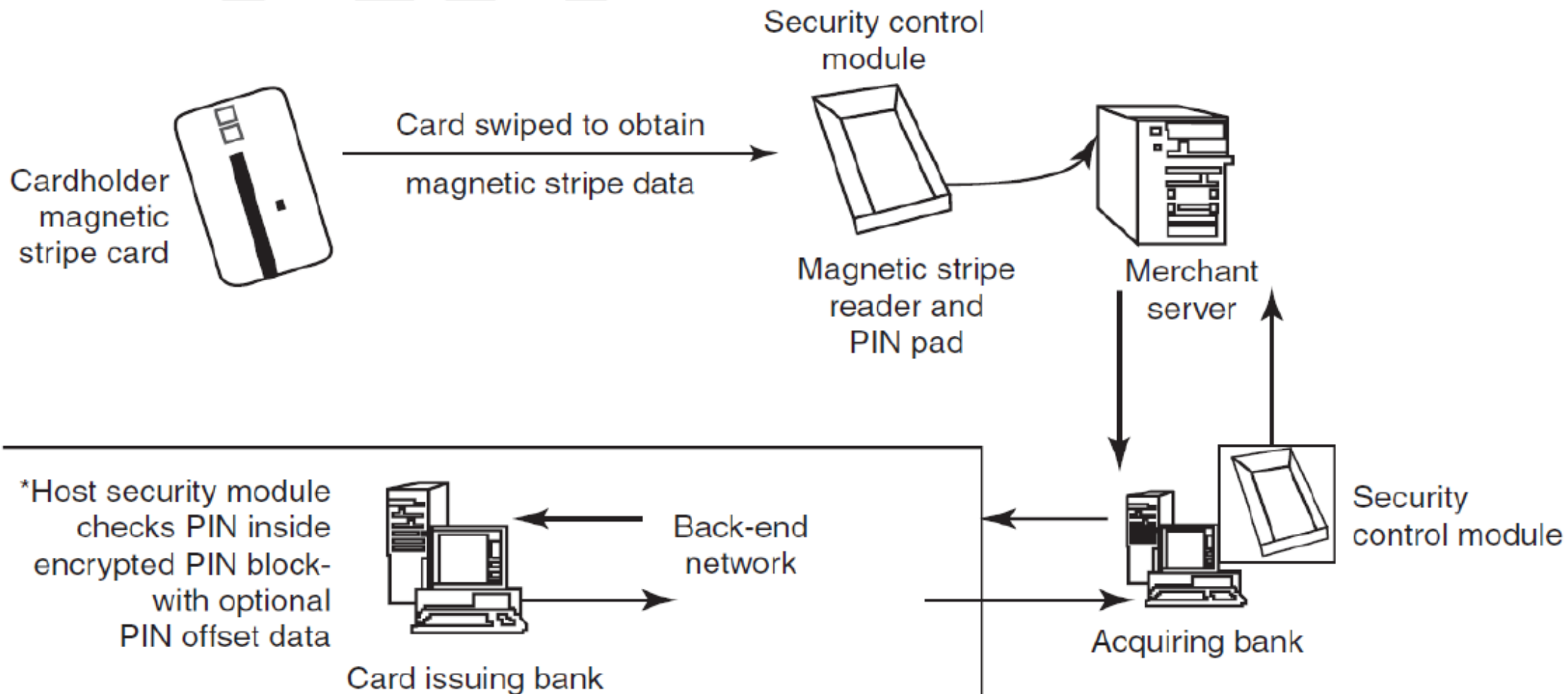
□ Mobile credit card transactions are now very common; new technologies combine lowcost mobile phone technologies with the capabilities of a point-of-sale (POS) terminal.

□ Today belongs to “mobile computing,” that is, anywhere anytime computing.

□ The developments in wireless technology have fuelled this new mode of working for white collar workers.

□ Wireless credit card processing is a very desirable system, because it allows businesses to process transactions from mobile locations quickly, efficiently and professionally.

□ It is most often used by businesses that operate mainly in a mobile environment.



- Credit card companies, normally, do a good job of helping consumers resolve identity (ID) theft problems once they occur. But they could reduce ID fraud even more if they give consumers better tools to monitor their accounts and limit high-risk transactions.

Tips to Prevent Credit Card Frauds

The current topic is about credit card frauds in mobile and wireless computing era, however, we would like to include these tips to prevent credit card frauds[8] caused due to individual ignorance about a few known facts.

Do's

1. Put your signature on the card immediately upon its receipt.
2. Make the photocopy of both the sides of your card and preserve it at a safe place to remember the card number, expiration date in case of loss of card.
3. Change the default personal identification number (PIN) received from the bank before doing any transaction.
4. Always carry the details about contact numbers of your bank in case of loss of your card.
5. Carry your cards in a separate pouch/card holder than your wallet.
6. Keep an eye on your card during the transaction, and ensure to get it back immediately.

7. Preserve all the receipts to compare with credit card invoice.
8. Reconcile your monthly invoice/statement with your receipts.
9. Report immediately any discrepancy observed in the monthly invoice/statement.
10. Destroy all the receipts after reconciling it with the monthly invoice/statement.
11. Inform your bank in advance, about any change in your contact details such as home address, cell phone number and E-Mail address.
12. Ensure the legitimacy of the website before providing any of your card details.
13. Report the loss of the card immediately in your bank and at the police station, if necessary.

Dont's

1. Store your card number and PINs in your cell.
2. Lend your cards to anyone.
3. Leave cards or transaction receipts lying around.
4. Sign a blank receipt (if the transaction details are not legible, ask for another receipt to ensure the amount instead of trusting the seller).
5. Write your card number/PIN on a postcard or the outside of an envelope.
6. Give out immediately your account number over the phone (unless you are calling to a company/ to your bank).
7. Destroy credit card receipts by simply dropping into garbage box/dustbin.

Types and Techniques of Credit Card Frauds

- Traditional Techniques
 - □ The traditional and the first type of credit card fraud is paper-based fraud - application fraud, wherein a criminal uses stolen or fake documents such as utility bills and bank statements that can build up useful personally Identifiable Information (PII) to open an account in someone else's name.
 - □ Application fraud can be divided into
 - 1. ID theft: Where an individual pretends to be someone else
 - 2. Financial fraud: Where an individual gives false information about his or her financial status to acquire credit. Illegal use of lost and stolen cards is another form of traditional technique. Stealing a credit card is either by pickpocket or from postal service before it reaches its final destination

- Modern Techniques
- □ Skimming is where the information held on either the magnetic strip on the back of the credit card or the data stored on the smart chip are copied from one card to another.
- □ Site cloning and false merchant sites on the Internet are becoming a popular method of fraud and to direct the users to such bogus/fake sites is called Phishing.
- □ Such sites are designed to get people to hand over their credit card details without realizing that they have been directed to a fake weblink /website (i.e., they have been scammed).

- 1. Triangulation: It is another method of credit card fraud and works in the fashion as
- explained further.
- • The criminal offers the goods with heavy discounted rates through a website designed and
- hosted by him, which appears to be legitimate merchandise website.
- • The customer registers on this website with his/her name, address, shipping address and
- valid credit card details.
- • The criminal orders the goods from a legitimate website with the help of stolen credit card details and supply shipping address that have been provided by the customer while
- registering on the criminal's website.
- • The goods are shipped to the customer and the transaction gets completed.
- • The criminal keeps on purchasing other goods using fraudulent credit card details of
- different customers till the criminal closes existing website and starts a new one.

- 2. Credit card generators: It is another modern technique - computer emulation software -
- that creates valid credit card numbers and expiry dates. The criminals highly rely on these
- generators to create valid credit cards. These are available for free download on the Internet.

Security Challenges Posed by Mobile Devices

- Mobility brings two main challenges to cybersecurity:
 - first, on the hand-held devices, information is being taken outside the physically controlled environment and
 - second remote access back to the protected environment is being granted.
- □ Perceptions of the organizations to these cybersecurity challenges are important in devising appropriate security operating procedure.
- □ As the number of mobile device users increases, two challenges are presented:
 - 1. at the device level called “microchallenges” and
 - 2. at the organizational level called “macrochallenges.”
- □ Some well-known technical challenges in mobile security are: managing the registry settings and configurations, authentication service security, cryptography security, Lightweight Directory Access Protocol (LDAP) security, remote access server (RAS) security, media player control security, networking application program interface (API) security, etc.

few antitheft software(s)

- Following are few antitheft software(s) available in the market:
- 1. GadgetTrak: <http://www.gadgettrak.com/products/mobile/>
- 2. Back2u: <http://www.bak2u.com/phonebakmobilephone.php>
- 3. Wavesecure: <https://www.wavesecure.com/>
- 4. F-Secure: <http://www.f-secure.com/>

Mishing

- Mishing is a combination of mobile and Phishing.
- □ Mishing attacks are attempted using mobile phone technology.
- □ M-Commerce is fast becoming a part of everyday life. If you use your mobile phone for
- purchasing goods/services and for banking, you could be more vulnerable to a Mishing
- scam.
- □ A typical Mishing attacker uses call termed as Vishing or message (SMS) known as
- Smishing.
- □ Attacker will pretend to be an employee from your bank or another organization and will
- claim a need for your personal details.
- □ Attackers are very creative and they would try to convince you with different reasons why
- they need this information from you.

- Vishing
- □ Vishing is the criminal practice of using social engineering over the telephone system,
- most often using features facilitated by VoIP, to gain access to personal and financial
- information from the public for the purpose of financial reward.
- □ The term is a combination of V - voice and Phishing.
- □ Vishing is usually used to steal credit card numbers or other related data used in ID theft
- schemes from individuals.
- □ The most profitable uses of the information gained through a Vishing attack include:
- 1. ID theft; 2. purchasing luxury goods and services; 3. transferring money/funds;
- 4. monitoring the victims' bank accounts; 5. making applications for loans and credit cards.

Smishing

- Smishing is a criminal offense conducted by using social engineering techniques similar to Phishing.
- □ The name is derived from "SMS PhISHING."
- □ SMS can be abused by using different methods and techniques other than information gathering under cybercrime.
- □ Smishing uses cell phone text messages to deliver a lure message to get the victim to reveal his/her PI.
- □ The popular technique to "hook" (method used to actually "capture" your information) the victim is either provide a phone number to force the victim to call or provide a website URL to force the victim to access the URL, wherein, the victim gets connected with bogus website (i.e., duplicate but fake site created by the criminal) and submits his/her PI.
- □ Smishing works in the similar pattern as Vishing.

Bluetooth hacking tools

- 1. BlueScanner: This tool enables to search for Bluetooth enable device and will try to
 - extract as much information as possible for each newly discovered device after
 - connecting it with the target.
- 2. BlueSniff: This is a GUI-based utility for fi nding discoverable and hidden Bluetooth
 - enabled devices.
- 3. BlueBugger: The buggers exploit the vulnerability of the device and access the images,
 - phonebook, messages and other personal information.
- 4. Bluesnarfer: If a Bluetooth of a device is switched ON, then Bluesnarfing makes it
 - possible to connect to the phone without alerting the owner and to gain access to
 - restricted portions of the stored data.
- 5. BlueDiving: Bluediving is testing Bluetooth penetration. It implements attacks like
 - Bluebug and BlueSnarf